0.1 A full gentle element in the extra CM coset

Statement E.58 (Proved). Relative pronilpotent lifting and the extra CM coset. *Let Γ be either the free profinite pants group $\widehat{F}(a,b)$, with peripherals $a, b, (ab)^{-1}$, or the actual modular group $\mathcal{B} = C_2 \amalg C_3 = \langle s, r \mid s^2 = r^3 = 1 \rangle$, with peripherals $s, r, (sr)^{-1}$. Let $U \triangleleft \Gamma$ be an open normal subgroup, and put*

$$D = \Gamma/U, \qquad R_U = R_{\mathrm{nil}}(U), \qquad \Gamma_{U,\mathrm{nil}} = \Gamma/R_U, \qquad \Gamma_{U,p} = \Gamma/O^p(U). \tag{E.58.1}$$

Here all residuals are closed, and $O^p(U)$ is the kernel of the maximal pro-p quotient of the actual subgroup U. Every continuous positive peripheral automorphism of $\Gamma_{U,\mathrm{nil}}$, with any specified three conjugators there, lifts to a continuous automorphism of Γ with those full conjugator coordinates retained. A specified peripheral whose old conjugator is one can be fixed literally in the lift. The exact local-source comparison is

$$\Gamma_{U,\mathrm{nil}} \xrightarrow{\sim} \prod_{p,\,D} \Gamma_{U,p}. \tag{E.58.2}$$

The product on the right consists of tuples with one common value in the specified finite quotient D.

In the conventions of E.1, set

$$\begin{gathered} \tau = sr, \qquad F = \ker(\mathcal{B} \to S_3), \qquad x = \tau^{-2}, \quad y = sxs, \qquad H = F'F^3, \\ q : \mathcal{B} \twoheadrightarrow Q = \mathrm{SL}_2(\mathbf{Z}/12)/\{\pm I\}, \\ q(s) = \left[\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\right], \quad q(r) = \left[\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}\right]. \end{gathered} \tag{E.58.3}$$

There is an actual full profinite gentle pair $(1, f)$, with

$$f \in R_{\mathrm{nil}}(H) \subset F', \tag{E.58.4}$$

whose normalized modular automorphism Φ has actual conjugators $A, \beta, V \in F$ satisfying

$$\begin{aligned} &\Phi(r) = ArA^{-1}, \qquad \Phi(s) = \beta s \beta^{-1}, \\ &\Phi(\tau) = V\tau V^{-1} = \tau, \qquad V = 1, \\ &q(A) = [5I], \qquad q(\beta) = q(V) = 1. \end{aligned} \tag{E.58.5}$$

Its two E.1 CM bits are $(1, 0)$. Thus $\widehat{\mathrm{GT}}_0$ reaches the extra CM coset, and Task E.2 has a positive answer. The construction is an actual full-source lift with these specified finite frames.

Proof. We first prove the relative source theorem, then construct its particular centralizer coordinate and calculate the CM signs.

The complete relative lifting theorem. Because U is normal, its characteristic residual R_U is normal in Γ, so all quotients in (E.58.1) are actual continuous quotients of the same source. Let α be the specified peripheral automorphism of $\Gamma_{U,\mathrm{nil}}$, with conjugators b_i. Work in the compact space Γ^3 of their prospective full lifts. Impose the peripheral product relation and generation in every finite quotient, and the values of the three conjugators in every finite quotient of $\Gamma_{U,\mathrm{nil}}$.

Take one actual finite joint source $q_0 : \Gamma \twoheadrightarrow E_0$ containing any prescribed finite collection of these constraints and the quotient D. Write

$$U_0 = q_0(U) = \ker(E_0 \to D), \qquad A_0 = \gamma_\infty(U_0), \qquad E_1 = E_0/A_0. \tag{E.58.6}$$

The subgroup A_0 is characteristic in U_0, hence normal in E_0, and $[E_0, A_0] = A_0$, because already $[U_0, A_0] = A_0$. Apply the finite scalar-cover construction D.50.5–17 to this actual extension. It gives an actual source quotient $E_1^\dagger$ central over E_1, including every subgroup count and every conjugator coordinate used there. Its image of U is a central extension of the nilpotent group U_0/A_0, and is therefore nilpotent. Consequently its source map kills R_U:

$$\Gamma \longrightarrow \Gamma_{U,\mathrm{nil}} \longrightarrow E_1^\dagger. \tag{E.58.7}$$

This proves the required factorization for this particular auxiliary quotient, including its original D-coordinate.

Precompose (E.58.7) by α. The new source map is onto and has the specified images of the b_i as literal peripheral conjugators. D.50.15–17 then supplies an epimorphism to E_0, with those conjugator values retained in every old factor. Every old relative finite constraint kills A_0, since the image of U in that factor is nilpotent. Thus this one finite epimorphism meets the entire chosen finite collection. In the modular case its source really is $C_2 \amalg C_3$: D.50.9 constructs the auxiliary maps with the two actual torsion relations, and the counted boundary classes retain these relations.

Compactness now gives three full conjugators and an endomorphism of Γ generating every finite quotient. Its image is compact, hence is all of Γ. The source is finitely generated, so the endomorphism is an automorphism by the Hopficity argument in D.50. This proves the full commuting quotient square and every specified conjugator value. If the conjugator of a retained peripheral is one, its full lift lies in R_U; conjugating the full automorphism by its inverse fixes that peripheral literally and leaves all the retained coordinates unchanged. This proof applies in particular to characteristic U; it also proves the stated normal-subgroup version with its specified finite base.

The exact product of the relative local sources. The actual map

$$U/R_{\mathrm{nil}}(U) \longrightarrow \prod_p U^{(p)} \tag{E.58.8}$$

is an isomorphism. Its kernel is the nilpotent residual, because a finite nilpotent group is the product of its Sylow groups. For surjectivity, every finite product of finite quotients at distinct primes is the actual source image: Goursat's common quotients have coprime orders and are trivial. The image in the infinite product is compact, so these finite surjections give the full product.

The kernel of (E.58.2) is therefore exactly R_U. For surjectivity, take a tuple with common value $d \in D$ and choose one $g \in \Gamma$ over d. Multiplying by the inverse of its actual tuple leaves an arbitrary element of $\prod_p U^{(p)}$; (E.58.8) supplies its source preimage. This proves (E.58.2) without choosing a splitting of any of its extensions.

In particular, local automorphisms inducing one common base automorphism, together with local peripheral conjugators having common D-values, define a peripheral automorphism of the displayed fibre product. The relative lifting theorem retains each of these full local maps and conjugators. This observation will be used with the identity automorphism in every component.

The actual finite Fermat source. The subgroup F is characteristic in $\mathcal{B}$. Indeed the six epimorphisms $C_2 * C_3 \to S_3$ send the two generators to a transposition and a three-cycle; they form one orbit under $\mathrm{Aut}(S_3)$ and have one kernel. The same assertion holds for the continuous maps from the profinite completion. Hence $H = F'F^3$ is characteristic as well. Its index is

$$[\mathcal{B} : H] = 6 \cdot 3^2 = 54, \qquad F/H \simeq (\mathbf{Z}/3)^2. \tag{E.58.9}$$

Let V_4 be the four matrices modulo four

$$I, \qquad \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}. \tag{E.58.10}$$

Every element of $q(F)$ has a unique representative modulo twelve whose reduction modulo four belongs to V_4. Multiplication of those representatives gives the actual isomorphism

$$q(F) \xrightarrow{\sim} V_4 \times \mathrm{SL}_2(\mathbf{F}_3), \qquad [M] \longmapsto (M \bmod 4, M \bmod 3). \tag{E.58.11}$$

The uniqueness follows from $-I \notin V_4$, and the Chinese remainder map proves injectivity. The literal x, y matrices generate all 96 elements of this product, as verified independently in the certificate below; this also identifies $q(F)$ with the kernel of reduction to $\mathrm{SL}_2(\mathbf{F}_2) = S_3$.

The derived subgroup of $\mathrm{SL}_2(\mathbf{F}_3)$ is its quaternion subgroup Q_8. One explicit pair of quaternion generators is

$$i = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \qquad j = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}; \qquad i^2 = j^2 = -I, \quad ij = -ji. \tag{E.58.12}$$

Their group has order eight and quotient of order three; the literal commutator and its conjugates generate it. Cubes kill that quotient, whereas cubing is the identity on V_4. Thus (E.58.11) gives exactly

$$q(H) = V_4 \times Q_8, \qquad |q(H)| = 32. \tag{E.58.13}$$

In particular the map q kills $O^2(H)$, so it factors through the actual source $\Gamma_2 = \mathcal{B}/O^2(H)$.

Set $D = \mathcal{B}/H$ and retain the actual joint image

$$\begin{aligned} J &= \mathrm{im}(\mathcal{B} \to Q \times D) = Q \times_{Q/q(H)} D, \\ |Q/q(H)| &= 18, \qquad |J| = 32 \cdot 54 = 1728. \end{aligned} \tag{E.58.14}$$

The map $D \to Q/q(H)$ is induced by q. To prove the fibre-product equality, lift the prescribed D-coordinate to $\mathcal{B}$; the discrepancy in its Q-coordinate lies in $q(H)$, and an actual element of H corrects it. The kernel over D is consequently exactly $q(H)$. This joint source also factors through Γ_2, and the kernel of

$$\Gamma_2 \twoheadrightarrow J \tag{E.58.15}$$

is a closed subgroup of $H^{(2)}$, hence is pro-two.

For an independent explicit model of D, use affine transformations of $\mathbf{F}_3^2$, written (L, v), with multiplication $(L, v)(M, w) = (LM, v + Lw)$. The images

$$s \longmapsto \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right), \qquad r \longmapsto \left(\begin{pmatrix} 0 & 2 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \tag{E.58.16}$$

satisfy the defining torsion relations. The literal x, y become translations by the two standard basis vectors. Their image has order 54, so its kernel is exactly H, by (E.58.9). The finite certificate enumerates the joint source through these same generator maps.

A specified relative centralizer element. The following computation is in the fixed matrices of (E.58.3):

$$\begin{aligned} q(x) &= \left[\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\right], \qquad q(y) = \left[\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}\right], \\ q([x,y]) &= \left[\begin{pmatrix} 1 & 8 \\ 8 & 5 \end{pmatrix}\right], \qquad z_0 := q([x,y]^2) = [5I]. \end{aligned} \tag{E.58.17}$$

Thus z_0 is central of order two in Q, and $z_0 \in q(H)$ has coordinates $(I, -I)$ under (E.58.11). The literal word $[x,y]^2 \in H$ shows that

$$(z_0, 1) \in C_J(r). \tag{E.58.18}$$

For clarity, here is the precise centralizer lifting argument. In a surjection $G \to E$ with pro-p kernel K, let $a \in G$ have finite order prime to p. Its image has the same order. Given any $e \in C_E(\bar{a})$, choose a lift g. The subgroups $\langle a \rangle$ and $\langle gag^{-1} \rangle$ are complements to K in the preimage of $\langle \bar{a} \rangle$. At each finite quotient, coprime complement conjugacy gives an element of the kernel conjugating them with the same generator image. The resulting closed conjugator sets in the compact group K are nonempty and have the finite intersection property. Their intersection gives $u \in K$ with $ugag^{-1}u^{-1} = a$. Thus ug centralizes a and lifts the prescribed e.

Apply this argument to (E.58.15), $p = 2$, $a = r$, and $e = (z_0, 1)$. The element r has order three in both groups, because the retained S_3 quotient detects that order. We obtain an actual element

$$c \in C_{\Gamma_2}(r) \cap H^{(2)}, \qquad c \longmapsto (z_0, 1) \text{ in } J. \tag{E.58.19}$$

More explicitly, it has the form $u[x,y]^2$, with $u \in \ker(\Gamma_2 \to J)$ supplied by the preceding compact conjugator construction. This specifies its actual maps, rather than replacing a finite centralizer element by an unproved full centralizer lift.

The full gentle automorphism and its exact frames. Use (E.58.2) with this $U = H$. Take the identity automorphism in every local component. Choose the peripheral conjugators

$$b_s = 1, \qquad b_\tau = 1, \qquad b_r = (c \text{ at } 2,\ 1 \text{ at every other prime}). \tag{E.58.20}$$

All components of b_r have value one in D, so this is an actual element of the indicated fibre product. It centralizes r in every component. Consequently these are literal peripheral conjugators for the identity automorphism of $\mathcal{B}/R_{\mathrm{nil}}(H)$.

The relative lifting theorem gives an automorphism of $\mathcal{B}$ with these full relative coordinates. Conjugate it by the inverse of its τ-conjugator, which lies in $R_{\mathrm{nil}}(H)$, to obtain Φ fixing τ. Its s- and r-conjugators β, A still have precisely the prescribed images $1, b_r$. Their values in D are one, so they are actual elements of $H \subset F$. Since q factors through the two-primary component, their finite values are $q(\beta) = 1$, $q(A) = z_0$, as in (E.58.5).

Put $f = s\Phi(s)$. The retained relative automorphism is the identity, so $f \in R_{\mathrm{nil}}(H) \subset H' \subset F'$. Exactly as in D.44, the torsion and product relations give

$$\begin{aligned} &\Phi(s) = sf, \qquad \Phi(r) = f^{-1}r, \qquad \Phi(x) = x, \qquad \Phi(y) = f^{-1}yf, \\ &f(y,x)f(x,y) = 1, \qquad f((xy)^{-1}, x)f(y, (xy)^{-1})f(x,y) = 1. \end{aligned} \tag{E.58.21}$$

The restriction to the characteristic subgroup F is an actual automorphism. These are the multiplier-one full gentle equations, so $(1, f) \in \widehat{\mathrm{GT}}_0$.

The full accessory frames here have no hidden pure centralizer ambiguity. Herfort–Ribes, Theorem B' (Section 4, printed page 159), proves for a free profinite product that the centralizer of a nonidentity factor element is its centralizer in that factor. Applied to the two specified factors of $\mathcal{B}$, this gives

$$C_{\mathcal{B}}(s) = \langle s \rangle, \qquad C_{\mathcal{B}}(r) = \langle r \rangle, \qquad C_{\mathcal{B}}(s) \cap F = C_{\mathcal{B}}(r) \cap F = 1. \tag{E.58.22}$$

The last equalities use the injective images of those factors in the specified S_3 quotient. Thus $\beta, A \in F$ are the unique pure conjugators for this normalized full automorphism. In particular the nontrivial relative r-frame is retained by an actual full automorphism; it has not been identified with a full centralizer element.[1]

The positive CM convention and the resulting coset. Use exactly E.1's representatives and coordinates $T = A^{-1}V$, $W = \beta^{-1}V$. Here $V = 1$, so

$$q(T) = [5I], \qquad q(W) = 1. \tag{E.58.23}$$

The unique representative of $[5I]$ whose mod-four value belongs to V_4 is $5I$ itself. Its inverse is again $5I$, with mod-three value $-I$. For the specified positive matrix $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$, the rational three-torsion line $\ker(R-I)$ modulo three is generated by $(1,2)^t$. The A^{-1} coordinate sends this vector to $(2,1)^t$, giving E.1's first bit one. The β^{-1} coordinate is the identity, giving its second bit zero. Thus the two restrictions compared in E.1 disagree. Through the precise finite-coordinate comparison of E.1 and LM7 D.25, this is the extra CM coset in Task E.2.

The finite centralizer calculation is essential for these coordinates: Φ induces the identity on Q, while its unique full r-accessory frame has value $[5I]$. The preceding construction and (E.58.22) prove exactly how those two statements hold for the same actual automorphism.

Comparison with the original homology shadows. For this construction one does not need a factorization of the entire G_6 source through Γ_2. In fact no such factorization exists for the specified maps. The element

$$x^6 = \tau^{-12} \in H \cap N_B \tag{E.58.24}$$

has primitive N_B-abelianization. To check this integrally, use the Bass–Serre graph of E.1. The chain of τ^{12} is

$$\sum_{j=0}^{11} \left(e_{q(\tau)^j q(s)} - e_{q(\tau)^{j+1}} \right). \tag{E.58.25}$$

It has zero boundary and 24 distinct nonzero edge coefficients, each 1 or -1. Hence it is primitive in the integral cycle lattice. Its image in $G_6 = B/(N_B' N_B^6)$ therefore has order six. But the image of H in Γ_2 is pro-two, so every finite image of its element has two-power order. This proves the asserted failure of that factorization through an explicit element.

The full automorphism just constructed nevertheless has an actual G_6 shadow. Indeed $q\Phi = q$, so Φ preserves N_B and its characteristic subgroup $N_B' N_B^6$. It therefore induces an actual automorphism of this specified G_6, and that shadow has the finite frames (E.58.5). E.9 excludes the particular earlier recorded representatives; the present construction gives a full-source point in the whole CM matrix fibre. It does not identify that point with one of those excluded representatives.

The independent standard-library program `bc_relative_cm_sources.py` enumerates the 576-element projective matrix source, its exact 96- and 32-element subgroup images, the affine 54-element quotient, and the full 1728-element joint source. It checks every product in (E.58.11), the literal word $[x,y]^2$, the positive signs $(1,0)$, and the primitive integral chain in (E.58.25). The infinite centralizer and peripheral lifts are the complete proofs above. □

References

[1] W. Herfort and L. Ribes. Torsion elements and centralizers in free products of profinite groups. J. reine angew. Math. 358 (1985), 155–161.